

Section	Ask	Points	What good looks like	What average looks like	What poor looks like	What nothing looks like	
		60	80-100%	60-80%	<60%	0%	% Weightage
Exploratory Data Analysis and Insights	- Problem definition, questions to be answered - Data background and contents - Univariate analysis - Bivariate analysis - Key meaningful observations on individual variables and the relationship between variables	8	1) Definition of the problem (as per given problem statement with additional views) 2) Observations on the shape of data, data types of various attributes, missing values, statistical summary. 3) Univariate Analysis (Boxplots and distribution plots for numerical data and bar plots for categorical data) 4) Bivariate Analysis (Correlation matrix, bivariate analysis for categorical and numerical variables with the target variable) 5) Comments on the range of attributes, outliers of various attributes 6) Insights from all the graphs plotted	1) Definition of the problem (as per given problem statement with additional views) 2) Observations on the shape of data, data types of various attributes, missing values, statistical summary. 3) Univariate Analysis (Boxplots and distribution plots for numerical data and bar plots for categorical data) 4) Bivariate Analysis (Correlation matrix) 5) Comments on the range of attributes, outliers of various attributes	1) Observations on the shape of data, data types of various attributes, missing values, statistical summary. 2) Univariate Analysis (Boxplots and distribution plots for numerical data and bar plots for categorical data) 3) Bivariate Analysis (Correlation matrix)	No code/report	12.50%
Data pre-processing	- Prepare the data for analysis - Feature Engineering - Missing value Treatment - Outlier Treatment - Ensure no data leakage among train-test and validation sets	5	1) Detect missing values in Education, Previous_year_rating and avg_training_score and treat them 2) Missing value treatment should be done after train, test and validation split 3) Oversampling and undersampling should be done on only train set 4) Test set should only be used at the end after getting a final model	1) Detect missing values in Education, Previous_year_rating and avg_training_score and treat them 2) Missing value treatment should be done after train, test and validation split 3) Oversampling and undersampling should be done on only train set	1) Detect missing values in Education, Previous_year_rating and avg_training_score and treat them 2) Missing value treatment should be done after train, test and validation split	No code/report	12.50%
Model building - Original Data	- Build 5 models (from decision trees, bagging, and boosting methods) - Comment on the model performance * You can choose NOT to build XGBoost if you are facing issues with the installation	6	1) Build 6 classification models and trained them using the original train data and Comment on the model performance	1) Build 3-6 classification models and trained them using the original train data	1) Build <3 classification models and trained them using the original train data	No code/report	7.50%
Model building - Oversampled data	- Oversample the train data - Build 5 models (from decision trees, bagging and boosting methods) - Comment on the model performance * You can choose NOT to build XGBoost if you are facing issues with the installation	6	1) Oversample the train data 2) Fit 5 tuned models on oversampled data 3) Check the performance of the 5 models 4) Comment on the model performance	1) Oversample the train data 2) Fit 3 tuned models on oversampled data	1) Oversample the train data 2) Fit <3 tuned models on oversampled data	No code/report	10.00%
Model building - undersampled data	- Undersample the train data - Build 5 models (from decision trees, bagging and boosting methods) - Comment on the model performance * You can choose NOT to build XGBoost if you are facing issues with the installation	6	1) Undersample the train data 2) Fit 5 tuned models on oversampled data 3) Check the performance of the 5 models 4) Comment on the model performance	1) Undersample the train data 2) Fit 3 tuned models on oversampled data	1) Undersample the train data 2) Fit <3 tuned models on oversampled data	No code/report	10.00%
Model Performance Improvement using Hyperparameter Tuning	- Choose models that might perform better after tuning (tune at least 3 models out of 15 built in the previous steps) - Provide proper reasoning for tuning that model - Tune the best 3 models obtained above using randomized search and metric of interest - Check the performance of 3 tuned models	13	1) Explained model evaluation criteria and selected 'Recall' as metric of interest 2) Choose 3 best models and provided the reason for the same 3) Tune the best 3 models obtained above using randomized search and metric of interest 4) Check the performance of 3 tuned models 5) Comment on the model performance	1) Explained model evaluation criteria and selected any other metric as metric of interest 2) Choose 3 best models 3) Tune the best 3 models obtained above using randomized search and metric of interest	1) Choose 3 best models 2) Tune the best 3 models obtained above using randomized search and metric of interest	No code/report	12.50%

Model Performances and Final Model Selection	- Compare the performance of tuned models - Choose the best model - Comment on the performance of the best model on the test set	4	1) Compare model performances of tuned models 2) Choose the best model 3) Find test performance 4) Comment on the model performance	1) Compare model performances of tuned models 2) Choose the best model 3) Find test performance	1) Compare model performances of tuned models 2) Choose the best model	No code/report	10.00%
Actionable Insights & Recommendations	- Write down insights from the analysis conducted - Provide actionable business recommendations	4	1) Insights (at least 4) 2) Recommendations (at least 2)	1) Recommendations provided but are not actionable.	1) Recommendations - irrelevant to the business problem	No code/report	7.50%
Notebook - Overall	- Structure and flow - Crispness - Visual appeal - Conclusion and Business Recommendations OR - Structure and flow - Well commented code - Conclusion and Business Recommendations	8	1) Clear structure and flow - everything sits well in a story 2) Crispness (Not too many words, just enough to keep the focus on key things/points) 3) Visual appeal (Use of charts, colors, diagrams, format, symmetry, informative visualizations that are easy to interpret) OR 1) Well structured notebook with a logical flow 2) Clean and well commented code	1) There is structure and flow but some bits are jumbled 2) Points are made but in too many words 3) Lesser charts, format is not the cleanest OR 1) There is structure and flow but some bits are missing 2) Some of the code is commented * If any section is missing and points have been deducted for the same in a previous section, then no points should be deducted in this section	1) No structure or flow 2) Only a few points are covered, story is not complete 3) Not many visuals used OR 1) no structure or flow 2) no comments in the code	No report OR No code	10.00%